DRIVEABILITY—INTERMITTENT ENGINE QUIT OR IDLE DIP—NO DIAGNOSTIC TROUBLE CODES (DTCS) PRESENT—VEHICLES EQUIPPED WITH 3.0L DURATEC ENGINE ONLY

Article No. 02-23-1

FORD: 2001-2003 ESCAPE

Article **02-11-6** is being republished in its entirety to update the Model Year and Calibration Information.

ISSUE

Some vehicles equipped with the 3.0L Duratec engine may exhibit an intermittent engine quit and restart condition. This is usually a one-time event during closed throttle deceleration with no Diagnostic Trouble Codes (DTCs) and no Malfunction Indicator Lamp (MIL). Due to the intermittent nature of the condition and the multiple potential causes of the condition, the complete bulletin checklist and all appropriate part replacements should be performed regardless of whether the condition can be duplicated by the technician. Otherwise, customers may experience the intermittent condition and be forced to return to the dealership. If the vehicle is no longer eligible for warranty coverage, discuss this service with the customer before performing.

ACTION

Addition to normal diagnostics, perform **ALL** of the following Driveability Checklist. Although the condition may not be possible to duplicate, it is recommended to **perform this bulletin checklist in its entirety to resolve the condition.**

SERVICE PROCEDURE

NOTE

THIS CONDITION MAY HAVE SEVERAL CAUSES, AND IT IS VERY IMPORTANT TO THOROUGHLY AND COMPLETELY PERFORM EACH STEP. IF EACH STEP IS NOT PERFORMED COMPLETELY, THE RESULT COULD BE AN INCOMPLETE OR REPEAT REPAIR.

NOTE

DUE TO THE INTERMITTENT NATURE OF THE CONDITION AND THE MULTIPLE POTENTIAL CAUSES OF THE CONDITION, THE COMPLETE BULLETIN CHECKLIST AND ALL APPROPRIATE PART REPLACEMENTS SHOULD BE PERFORMED REGARDLESS OF WHETHER THE CONDITION CAN BE DUPLICATED BY THE TECHNICIAN.

Please use the following conditions for all tests described below unless stated otherwise:

- Transmission in Park
- Engine at idle at approximately 750 RPM
- Engine temperature should be at least 190° F (88° C)
- All accessories and the engine cooling fan should be off

1. Determine if the Evaporative Vapor Management (EVAPVM) duty cycle is operating properly. If EVAPVM is functioning correctly, the duty cycle should increase to 84-100% with the FTP decreasing to approximately 2.2 volts and then reset back to 0% duty cycle while FTP holds at approximately 2.6 volts. If the duty cycle does not increase within 5 minutes, turn on the headlights and the AC with the blower on high. The duty cycle should start increasing within 5-10 minutes. Do not replace the EVAPVM valve if the duty cycle functions correctly. If the duty cycle stops increasing and remains at 95-100% while FTP holds at approximately 2.6 volts, replace the EVAPVM valve with part number YL8Z-9C915-AA. Verify corrective action then proceed to Step 2.

NOTE

2003 VEHICLES BUILT IN LATE 2002 CALENDAR YEAR OR AFTER WILL HAVE A REVISED EVAPORATIVE EMISSIONS SYSTEM THAT REMOVES THE CHECK VALVE. IF THE VEHICLE IS NOT EQUIPPED WITH A CHECK VALVE, IT IS NOT NECESSARY TO PERFORM STEP 2 OF THIS PROCEDURE.

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2. Disconnect the vent line in the evaporative emissions system from the check valve side (for reference check valve part # is YL8U-9C915-AB). This connection is located just forward of the evaporative emissions canister assembly, underneath the vehicle in the area of the driver side rear seat. Using shop air, blow the vent line from the check valve side forward (towards the brake booster). In the past, spiders have been known to construct webs in vent lines so handle with caution. Possible obstructions in the vent line can prevent the evaporative emissions system from purging properly, and in some cases, can cause the condition to occur. Verify corrective action then proceed to Step 3.

3. Inspect the Idle Air Control (IAC) Valve. If the valve is not part number 1L8E-9F715-AA, replace with part number 1L8Z-9F715-AA which will also require latest calibration level given in Step 4 or 5.

NOTE
ENGINE RPM WILL SLOWLY RAMP UP. IT IS EXTREMELY IMPORTANT TO CHECK THE IAC DUTY CYCLE WHEN THE RPM IS AT 750 RPM. EVEN 800 RPM IS TOO HIGH FOR CHECKING IAC DUTY CYCLE UNDER THESE CONDITIONS. IF RPM IS OVER 750 RPM, MOMENTARILY OPENING AND CLOSING THE THROTTLE WILL LOWER THE RPM.

Verify that IAC duty cycle is between 32-40% with no purge flow (EVAPVM duty cycle is 0%) and fuel trims (SHRTFT1, SHRTFT2, LONGFT1, LONGFT2) are less than 15%. If IAC duty cycle is within specification proceed to Step 4 now. If IAC duty cycle is out of specification, replace the throttle body with part number 2L8Z-9E926-AB. If the fuel trims are above 15%, disconnect the Mass Air Flow Sensor (MAF) and recheck the fuel trims. If the fuel trims drop to below 15%, replace the MAF sensor with part number 1L2Z-12B579-BARM. If fuel trims stay above 15%, check for vacuum leaks and check the fuel system. Verify corrective action then proceed to Step 4.

4. **For vehicles sold in the U.S. and Canada perform the following:** Reprogram PCM with WDS. Some 2001 model year PCMs cannot be reprogrammed and must be replaced. This is determined by the MPC # located in upper left corner of the barcode on the PCM. If the PCM is an MPC 160, then replace with part 1U7Z-12A650-AXD. If the PCM is MPC 161, then just reprogram with WDS. Verify latest calibration was successfully reprogrammed. WDS should show latest calibration level as 1U7A-12A650-AXD for 2001, 2U7A-12A650-CZB for 2002 or 3L8A-12A650-BC for 2003. Note that some early 2002 vehicles may have the 2001 calibration. Proceed to Step 5.

5. **For vehicles sold in Mexico perform the following:** Reprogram PCM with WDS. Some 2001 model year PCMs cannot be reprogrammed and must be replaced. This is determined by the MPC # located in upper left corner of the barcode on the PCM. If the PCM is an MPC 160, then replace with part 1U7Z-12A650-AZD. If the PCM is MPC 161, then just reprogram with WDS. Verify latest calibration was successfully reprogrammed. WDS should show latest calibration level as WDS should show latest calibration level as 1U7A-12A650-AZD for 2001MY, 2U7A-12A650-CPB for 2002MY or 3L8A-12A650-CC for 2003. Note that some early 2002 vehicles may have the 2001 calibration. Proceed to Step 6.

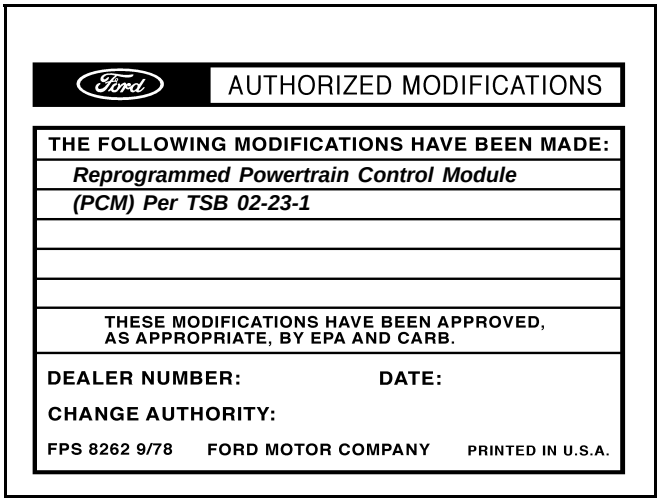
6. If the Electronic Engine Control (EEC) relay has stamped lettering, proceed to Step 7 now. If the EEC relay has white lettering printed on the top surface, replace with a new Hella service relay that is all black and has stamped lettering on the top surface. Both the new Hella service relay and the old relay have the same part number (FOAZ-14N089-A). Make sure the relay you are installing has stamped lettering. For location, use 2001 Wiring Diagram sections 303-07B-00-1 Connector C1016, 700-06-00-37 Battery Junction Box. Proceed to Step 7.

7. For 2001 and 2002 vehicles, inspect the DPFE sensor part number. If DPFE sensor is part number 2F1E-9J460-AA, proceed to Step 8 now. If the DPFE sensor is part number YF1E-9J460-AD, check for a white dot on the sensor housing (Note: White dot can be anywhere on housing). If there is a white dot, proceed to Step 8 now. If there is not a white dot, replace the DPFE with part 2F1Z-9J460-AA. Proceed to Step 8.
8. Ensure the Mass Air Flow (MAF) sensor gasket is properly installed and not blocking the air stream by disconnecting the airbox and looking inside the airbox towards the MAF sensor. If gasket is damaged, replace with part YL8Z-9E931-CA. Proceed to Step 9.
9. Verify the PCM harness integrity by removing the module from the COWL and moving the PCM harness around while the engine is running. If any abnormalities are observed, repair/replace the harness. Proceed to Step 10.
10. Inform the customer that significant weight (approximately 9 oz. or more) hanging from the key ring while the keys are in the ignition may move the ignition cylinder out of the Run position and stop the engine. When this occurs, all gauges immediately shut off (fuel reads "E", tachometer goes to zero, speedometer goes to zero immediately). It is recommended that the customer attach fewer keys to the key ring that retains the vehicle ignition key.

PCM CALIBRATION INFORMATION

Application	Old Part Number (-12A650-)	Tear Tag	New Part Number (-12A650-)	Old Calibration	New Calibration	NGS/WDS Qualifier
2001 3.0L - Escape	1U7A-AXB	ATF3	1U7Z-AXD	0M11A30512	0M11A30512	WDS B21.3 Release or Later
2001 3.0L - Escape	1U7A-AZB	ESG3	1U7Z-AZD	0M11B30512	0M11B30512	WDS B21.3 Release or Later
2002 3.0L - Escape	2U7A-CZA	NSF1	2U7Z-CZB	2M11A30510	2M11A30510	WDS B21.3 Release or Later
2002 3.0L - Escape	2U7A-CPA	PVN1	2U7Z-CPB	2M11B30510	2M11B30510	WDS B21.3 Release or Later
2003 3.0L - Escape	3L8A-BA	BAR2	3L8Z-BC	3M11A30510	3M11A30511	WDS B21.3 Release or Later
2003 3.0L - Escape	3L8A-CA	CAS2	3L8Z-CC	3M11B30510	3M11B30511	WDS B21.3 Release or Later

Obtain an Authorized Modifications Decal (FPS 8262 - obtainable through DOES II, 25/pkg) and list the date, dealer number, and summary of modifications performed. Select a prominent place adjacent to the Vehicle Emission Control Information Decal suitable for installing the Authorized Modifications Decal. Clean the area, install the decal, and cover it with a clear plastic decal shield.



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PART NUMBER	PART NAME
1L8Z-9F715-AA	IAC - Idle Air Control Valve
2L8Z-9E926-AB	Throttle Body
1L2Z-12B579-BARM	MAF - Mass Air Flow Sensor
YL8Z-9E931-CA	Gasket - Mass Air Flow Sensor
FOAZ-14N089-A	EEC Relay
2F1Z-9J460-AA	DPFE Sensor
YL8Z-9C915-AA	EVAPVM Valve

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 02-11-6
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 607000, 607400, 607500, 607600, 607700, 611000, 611500, 614000, 614500, 614600

NOTE: The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by "do-it-yourselfers". Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.